

Using uncertainty quantification to improve reliability of video-based skill assessment metrics in central venous catheterization

Catherine Austin, Rebecca Hisey, Olivia O'Driscoll, Tamas Ungi, Gabor Fichtinger

Laboratory for Percutaneous Surgery, School of Computing, Queen's University, Kingston, Canada

ABSTRACT

Purpose: Computed-based skill assessment relies on accurate metrics to provide comprehensive feedback to trainees. Improving the accuracy of video-based metrics computed using object detection is generally done by improving the performance of the object detection network, however increasing its performance requires resources that cannot always be obtained. This study aims to improve the accuracy of metrics in central venous catheterization without requiring a high performing object detection network by removing false positive predictions identified using uncertainty quantification. **Methods:** The uncertainty for each bounding box was calculated using an entropy equation. The uncertainties were then compared to an uncertainty threshold computed using the optimal point of a Receiver Operating Characteristic curve. Predictions were removed if the uncertainty fell below the predefined threshold. 50 videos were recorded and annotated with ground truth bounding boxes. These bounding boxes were used to train an object detection network, which was used to produce predictive bounding boxes for the test set. This method was evaluated by computing metrics for the predictive bounding boxes with and without having false positives removed and comparing them to ground truth labels using a Pearson Correlation. **Results:** The Pearson Correlations for the baseline comparisons and the comparisons made using the results calculated using false positive removal were 0.922 and 0.816 for syringe path lengths, 0.753 and 0.510 for ultrasound path lengths, 0.831 and 0.489 for ultrasound usage times, and 0.857 and 0.805 for syringe usage times. **Conclusions:** This method consistently reduced inflated metrics, making it promising for improving metric accuracy.

Keywords: Object detection, computer-assisted skill assessment, central venous catheterization

1. INTRODUCTION

Computer-based assessment of surgical skill relies on the analysis of several metrics to produce an accurate evaluation of a user's performance. One such computed-based assessment platform is Central Line Tutor developed by Hisey et al, which provides automatic skill assessment for central venous catheterization (CVC) [1]. Central Line Tutor relies on metrics such as path length and usage time to determine user skill [2]. Consistently low values of these metrics are associated with a higher skill level. Shorter path lengths and usage times demonstrate that a user can make tool movements that are both quick and smooth, which results in fewer complications experienced during and after a procedure [3]. Therefore, Central Line Tutor focuses on minimizing these metrics to improve trainee skill. To ensure the platform is accurately determining whether these metric values are decreasing, Central Line Tutor requires a system that accurately tracks the movement of the tools in its setup.

Traditionally, tracking for computer-based assessment platforms like Central Line Tutor is performed using external tracking systems that are both bulky and expensive. Video-based skill assessment using object detection to track tools is more convenient to use and cost-effective, however it is also less accurate than systems relying on external tracking. This can make the use of video-based skill assessment problematic, as the reduced accuracy of the system can lead to inaccurate evaluation of the user's skill. Improving the accuracy of metrics is generally done by acquiring large datasets for training or by augmenting data, which improves the overall performance of an object detection network. These approaches are often reliant on an ability to collect large amounts of data, which is not always feasible, particularly during the development of new skill assessment platforms that require the collection of entirely novel data. This suggests a need for a method that can improve the performance of video-based skill assessment without needing to acquire more data.

Identifying and removing false predictions may be one method of improving the reliability of metrics by mitigating the effect of errors that are inherent in any deep learning approach. When an object detection network falsely predicts either the presence or the absence of a tool it results in a skewing of metrics that either artificially inflates or deflates a user's computed skill level. In the case of Central Line Tutor, the skewing of metrics is especially prevalent for false positive

predictions, as false positives often inflate metrics such as path length and usage time, which leads to an inaccurately low level of user skill relative to performance. False positives in video-based tracking are also more difficult to overlook compared to false negative predictions, which can often be identified based on surrounding frames. To identify a false positive prediction, the level of confidence for each prediction made by the object detection model must be found. Predicted confidences can be calculated using various uncertainty quantification methods. Unfortunately, most uncertainty quantification methods are very computationally costly to implement and are therefore unable to assist with real-time skill evaluation. For this reason, we use an entropy equation to quantify the level of uncertainty present in each prediction to allow for our method to deliver results to a trainee in a timely manner. By evaluating the predictions made by the object detection network and removing those that were likely false positives, we aim to show that inflated metrics can be reduced without having to rely on an object detection network with a high performance.

2. METHODS

The uncertainty for each tool in the frame of a video was obtained using the Shannon entropy equation shown in Equation 1 [4]. The values used for calculating uncertainty were taken from the second last layer for the network, as the confidence values generated by the final softmax layer have proven to be unreliable for measuring uncertainty [5]. Then, a threshold of uncertainty needed to be established for each tool. If the model deems that the uncertainty of a prediction for the corresponding tool exceeds the threshold, the prediction can be deemed a false positive and be overlooked. This threshold was defined as the optimal point of a Receiver Operating Characteristic (ROC) curve.

$$H(X)_{\text{discr}} = -\sum_{x \in \Omega} p(x) \cdot \log_2(p(x)) \quad (1)$$

10 participants were each recorded performing CVC five times each to collect a total of 50 video recordings. The recordings were obtained from five anesthesiologists who were certified to practice without supervision and five medical students with no prior experience performing CVC. The videos were broken down into individual frames using the Deep Learn Live extension of 3D Slicer (<https://github.com/SlicerIGT/aigt/tree/master/DeepLearnLive>). Each of the frames was annotated with a bounding box location of the syringe and the phantom.

Metrics for path length and usage time were calculated using the ground truth labels for each video. The ground truth data was then used to train a You Only Look Once (YOLO) object detection network. The predictions made by the network were also used to calculate the same metrics as those used for the ground truth data. The two sets of metrics were used to establish a baseline comparison between ground truth bounding boxes and those made using object detection. The predictive bounding boxes generated by YOLO were then evaluated using the previously described entropy equation, and the results that were determined to be false positives were removed. Metrics were then calculated for the modified set of predicted bounding boxes. The metrics for this set were compared to the baseline differences in performance for the ground truth and predicted bounding boxes using a Pearson Correlation.

3. RESULTS

Table 1 shows the uncertainty thresholds calculated from the ROC curve for each tool.

Table 1. The uncertainty thresholds for each tool.

Tool	Threshold
anesthetic	-278.86
catheter	-58.848
dilator	-6.568
guidewire	-932.612
guidewire casing	-107.266
phantom	-146.892
scalpel	-6.568
syringe	-133.073
ultrasound	-123.001

Figure 1 shows a comparison of computed and ground truth path lengths for the syringe.

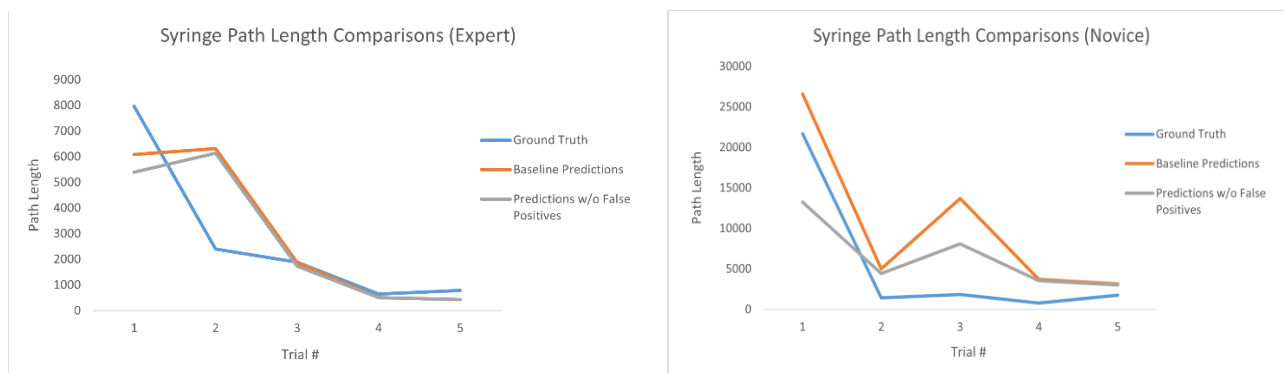


Figure 1. *Left.* A comparison of syringe path lengths for ground truth and computed bounding boxes with and without false positive removal for an expert participant. *Right.* A comparison of syringe path lengths for ground truth and computed bounding boxes with and without false positive removal for a novice participant.

Figure 2 shows a comparison of computed and ground truth usage times for the ultrasound.

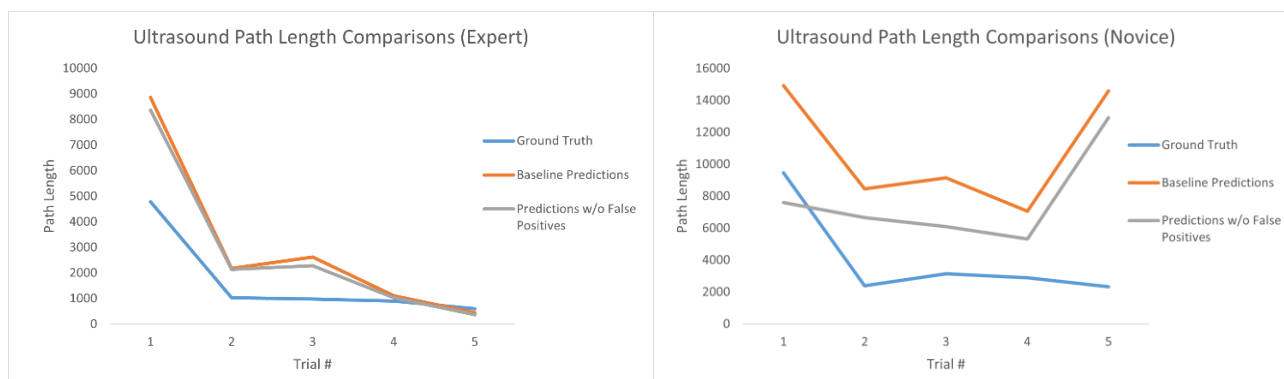


Figure 2. *Left.* A comparison of ultrasound path lengths for ground truth and computed bounding boxes with and without false positive removal for an expert participant. *Right.* A comparison of ultrasound path lengths for ground truth and computed bounding boxes with and without false positive removal for a novice participant.

Figure 3 shows a comparison of computed and ground truth usage times for the syringe.

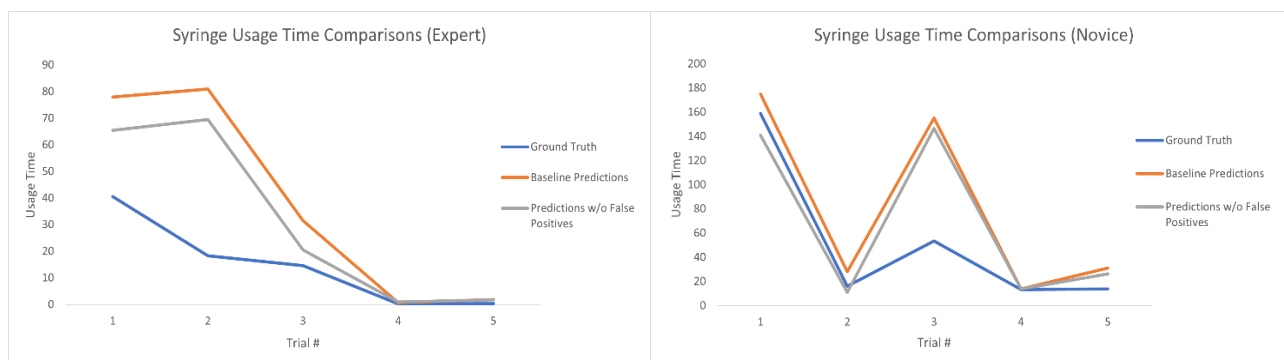


Figure 3. *Left.* A comparison of syringe usage times for ground truth and computed bounding boxes with and without false positive removal for an expert participant. *Right.* A comparison of syringe usage times for ground truth and computed bounding boxes with and without false positive removal for a novice participant.

Figure 4 shows a comparison of computed and ground truth usage times for the ultrasound.

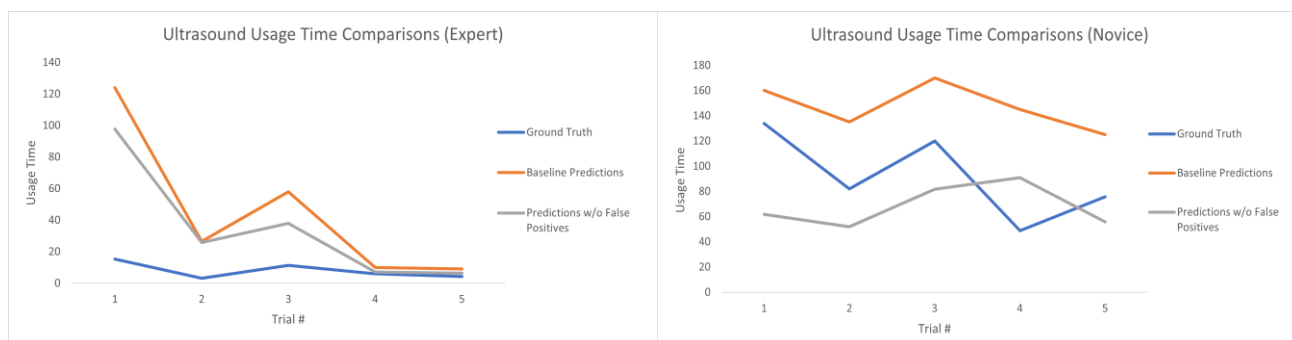


Figure 4. *Left.* A comparison of ultrasound usage times for ground truth and computed bounding boxes with and without false positive removal for an expert participant. *Right.* A comparison of ultrasound usage times for ground truth and computed bounding boxes with and without false positive removal for a novice participant.

The Pearson Correlation for the syringe path lengths was 0.922 for the baseline comparison compared to 0.816 for the comparison made using the results that had false positives removed. The respective Pearson Correlations for the ultrasound path lengths were 0.753 and 0.510. The Pearson Correlations for usage times were 0.831 for the baseline and 0.489 when using false positive removal for the ultrasound, and 0.857 for the baseline and 0.805 when using false positive removal for the syringe.

NEW OR BREAKTHROUGH WORK TO BE PRESENTED

The goal of this study was to find a method for improving the reliability of video-based skill assessment metrics that doesn't require the use of a highly accurate object detection network. It achieves this by presenting a method for reducing inflated metrics by identifying and eliminating false positive predictions produced by a network. Using this method, we show that removing the lower confidence predictions of an already trained network can produce significant improvements in the accuracy of metrics computed using object detection without needing to improve the performance of the network itself. This shows that metrics produced using object detection can be reliably computed without using the resources required to train a network with a high level of performance.

DISCUSSION AND CONCLUSIONS

The inflation of path length and usage times were consistently reduced when using false positive removal, indicating that the method presented in this study is an effective approach to reducing inflated metrics. The Pearson Correlations for the comparisons between the ground truth and metrics calculated using false positive removal were significant for all results except for ultrasound usage time. However, the Pearson Correlations for path length and usage time both implied the baseline comparison of predicted and ground truth metrics had a stronger positive correlation. This does not indicate that the individual values for path length and usage time are more accurate when using the method to calculate the baseline, as the metrics calculated using false positive removal are often closer in value to the ground truth metrics. Instead, it seems to show that the metrics computed with the traditional method are better at showing the general trend in user skill compared to the metrics computed with false positive removal. The decreasing of the Pearson Correlation values when using false positive removal could be attributed to the small number of trials the Pearson Correlation was performed on. For this reason, we will increase our sample size in future work to further investigate the effectiveness of the false positive removal method in regard to predicting general trends.

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